Lander Optimal Control

P.21-23 Robot Motion, Simulation, Optimization

Suraj Dayma July 04, 2026

1. Lander

2D point mass drone / lander with boundary conditions, umax, z >= 0 constraints.

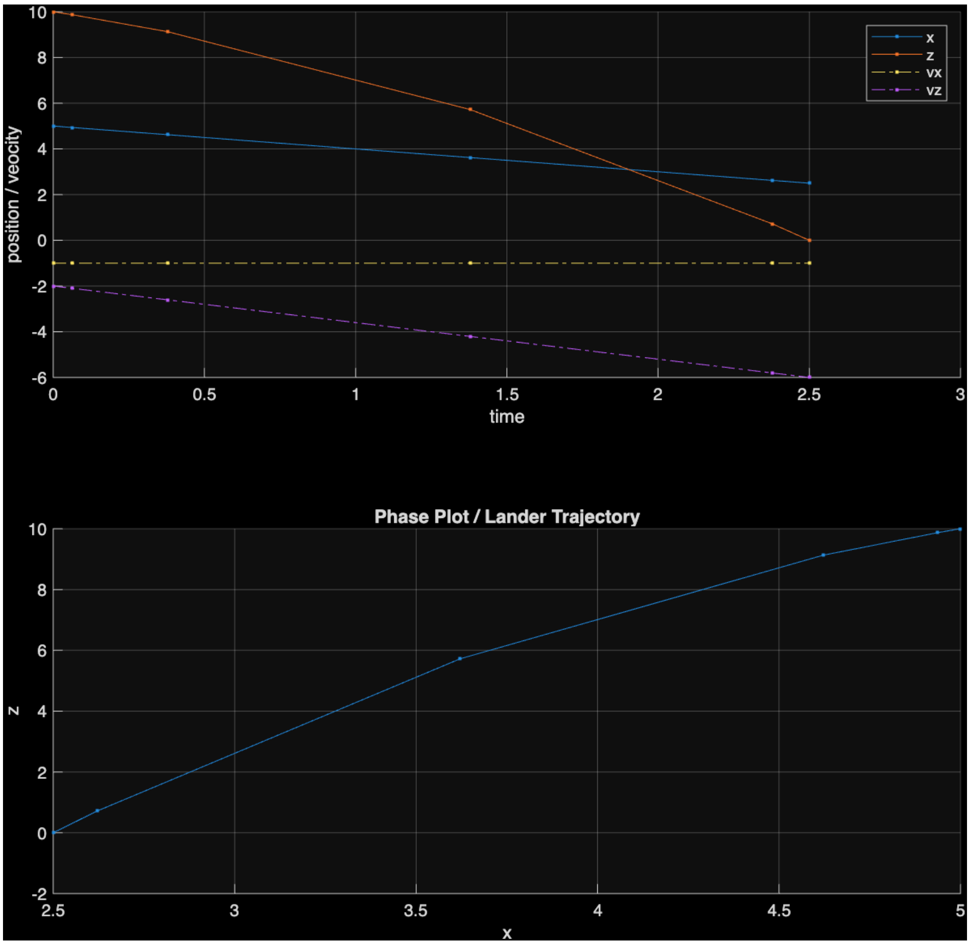


Figure 1: Sample trajectory without any control applied

2. Problem 21

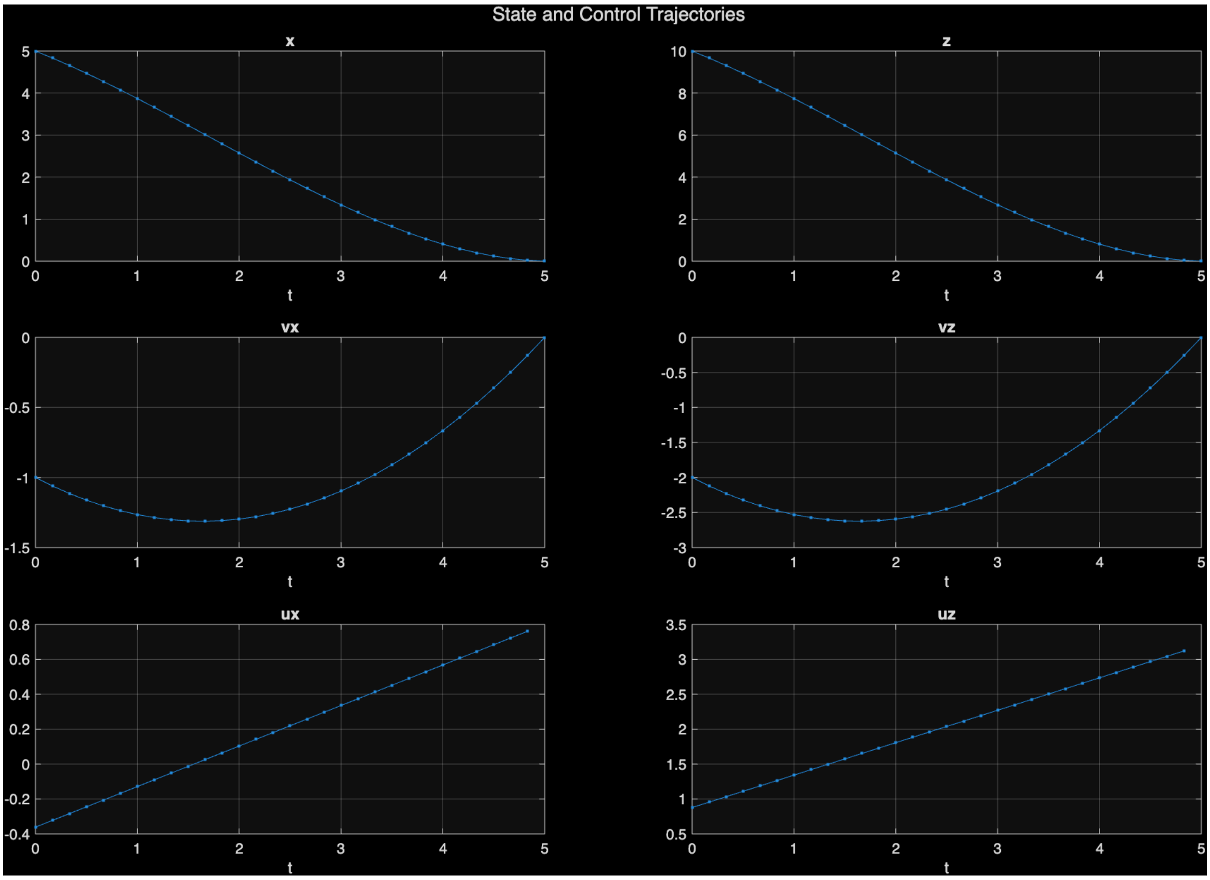
- I used a R matrix of [5, 1] where 5 is the cost to go for ux and 1 is the cost to go for uz, so control in x direction is more expensive.

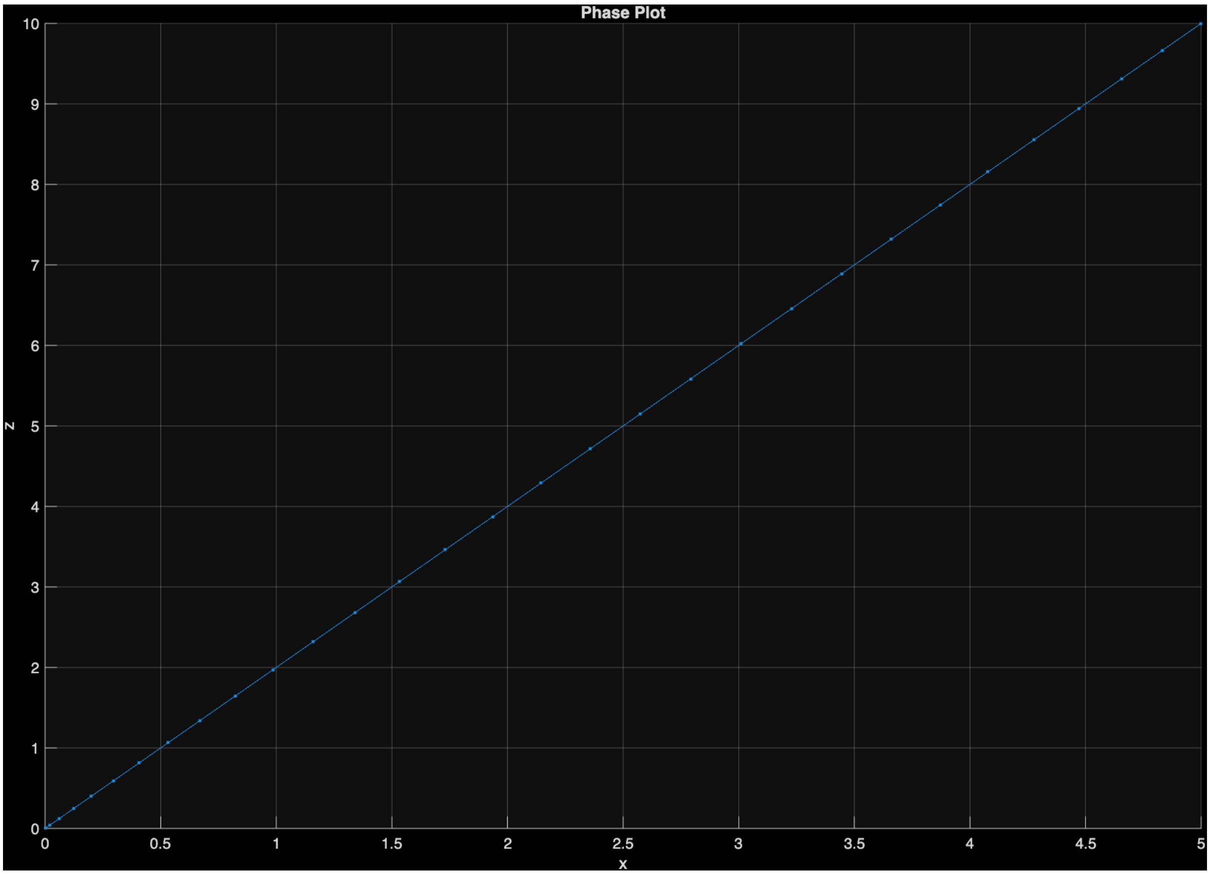
2.1. Optimal Trajectory

With the following parameters

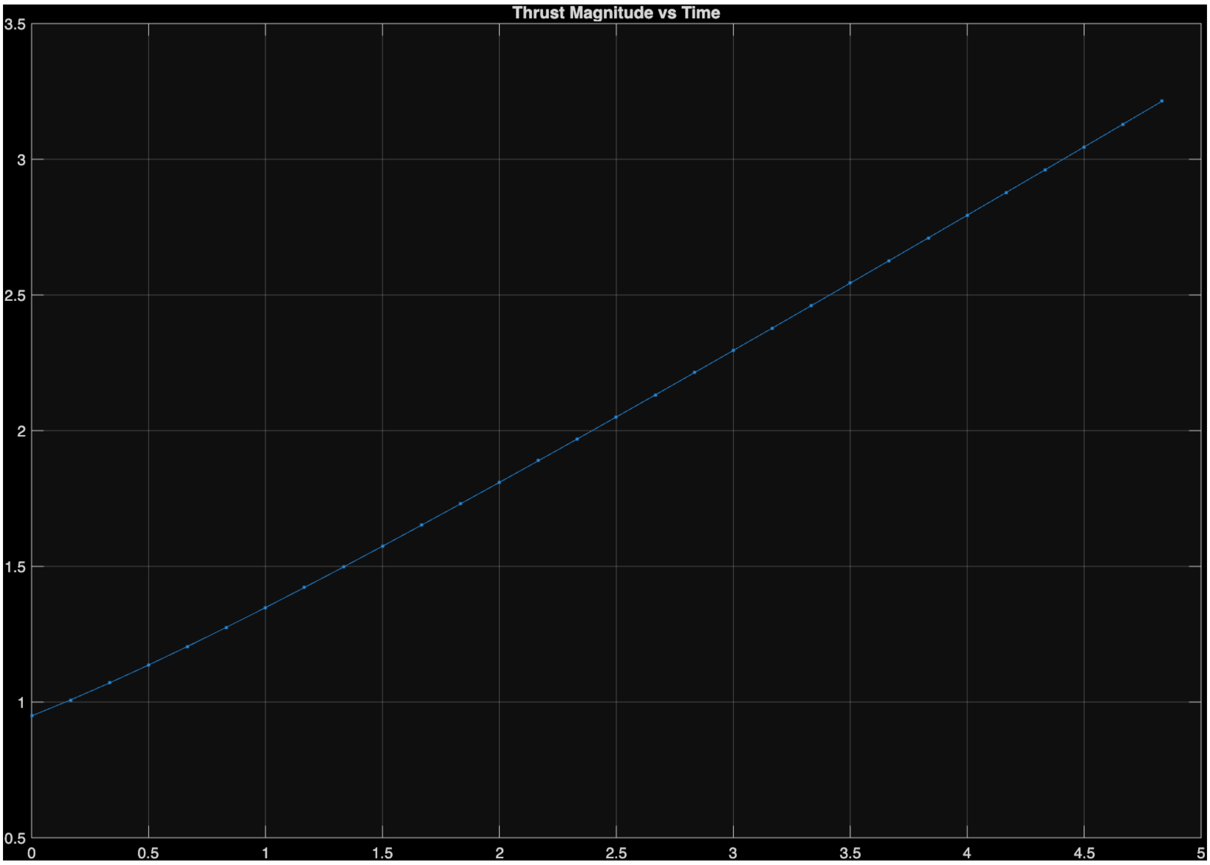
```
N = 30
R = diag([1, 1])
umax = 20;

% 4 variables: x,z,vx,vz which will have (N+1) points
% 2 control variables: ux, uz which will have N points
% Indexing: x  - 1          : N+1
%           z  - 1(N+1) + 1 : 2(N+1)
%           vx - 2(N+1) + 1 : 3(N+1)
%           vz - 3(N+1) + 1 : 4(N+1)
%           ux - 4(N+1) + 1 : 4(N+1) + N
%           vz - 4(N+1) + N + 1 : 4(N+1) + 2N
```





Phase Plot
z
x
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
1
2
3
4
5
6
7
8
9
10



Thrust Magnitude vs Time
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0.5
1
1.5
2
2.5
3
3.5

2.2. Feasibility Region

For same vx0 = −1, vz0 = −2:

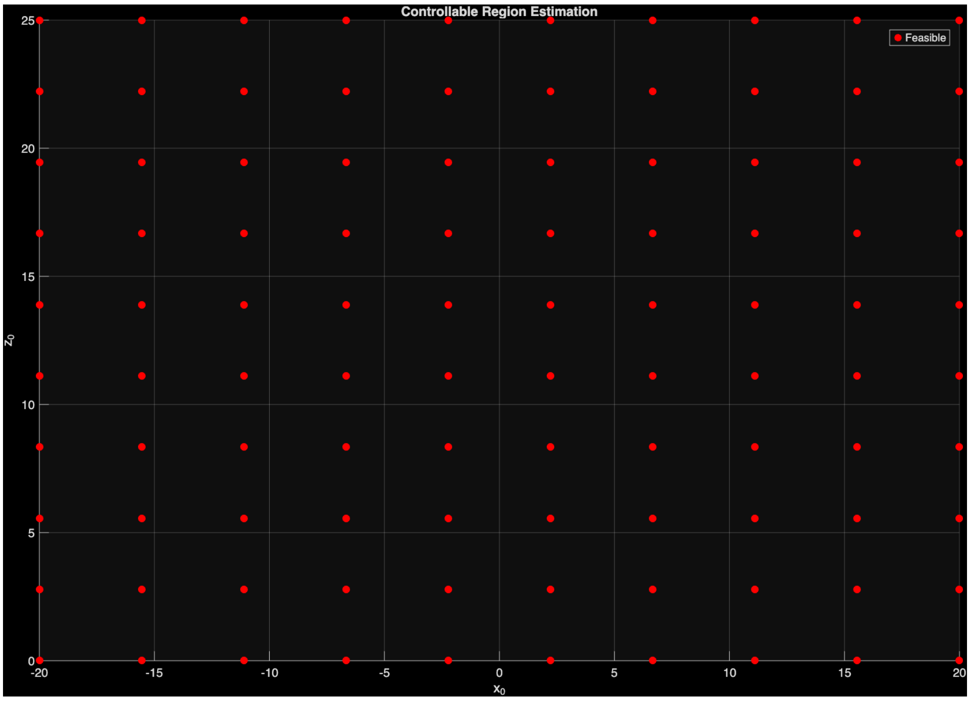


Figure 5: Feasibility Region for umax = 1 and T = 5 | Same for (umax, T) = (1,10), (2, 5)

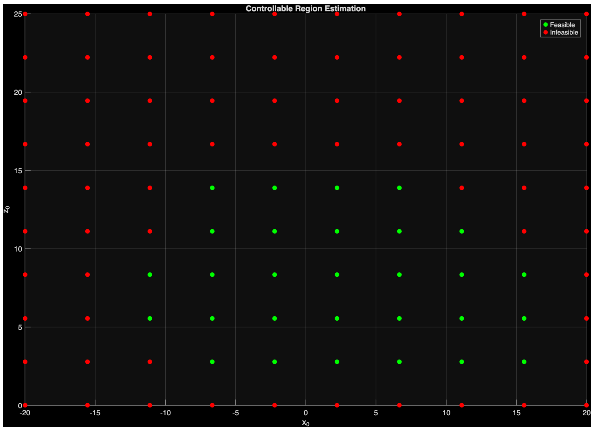


Figure 6: umax = 3 and T = 5

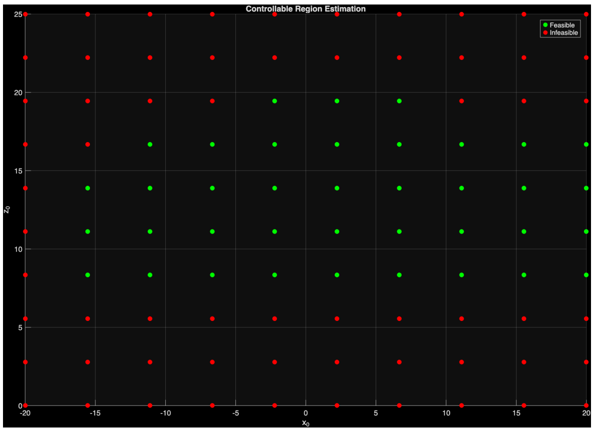


Figure 7: umax = 2 and T = 10

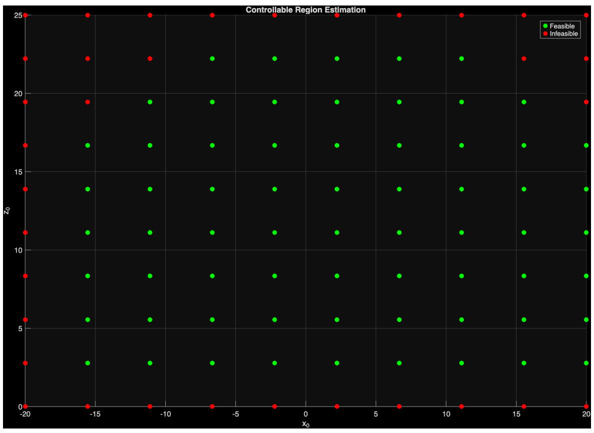


Figure 8: umax = 4 and T = 5

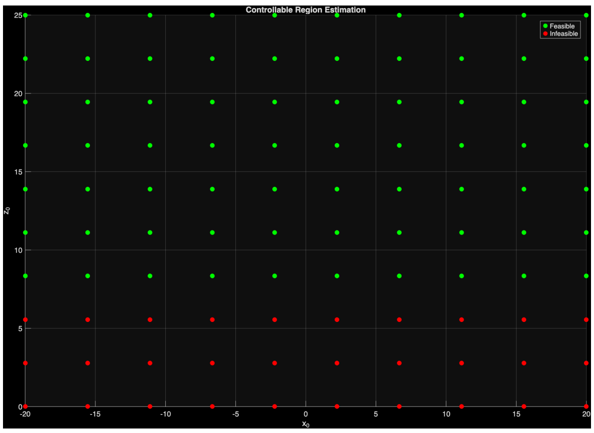
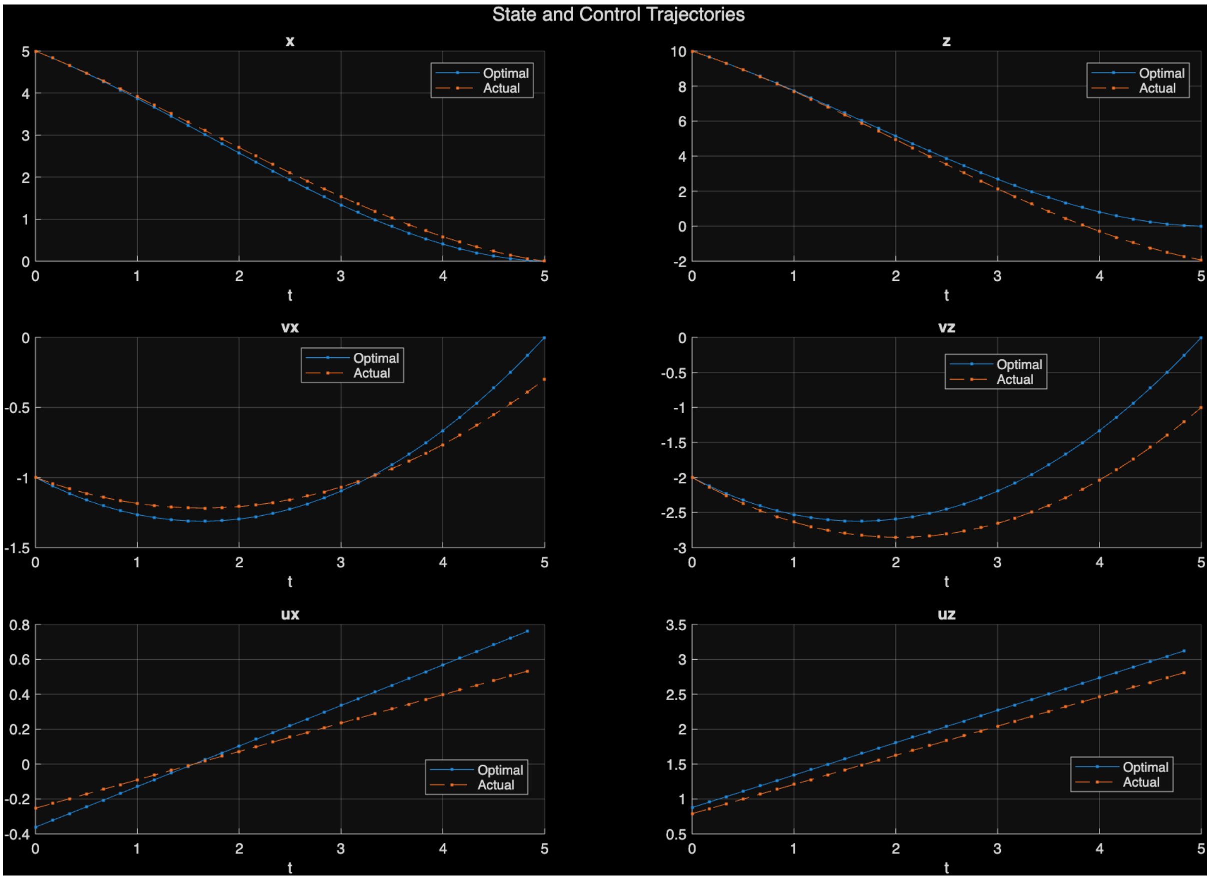
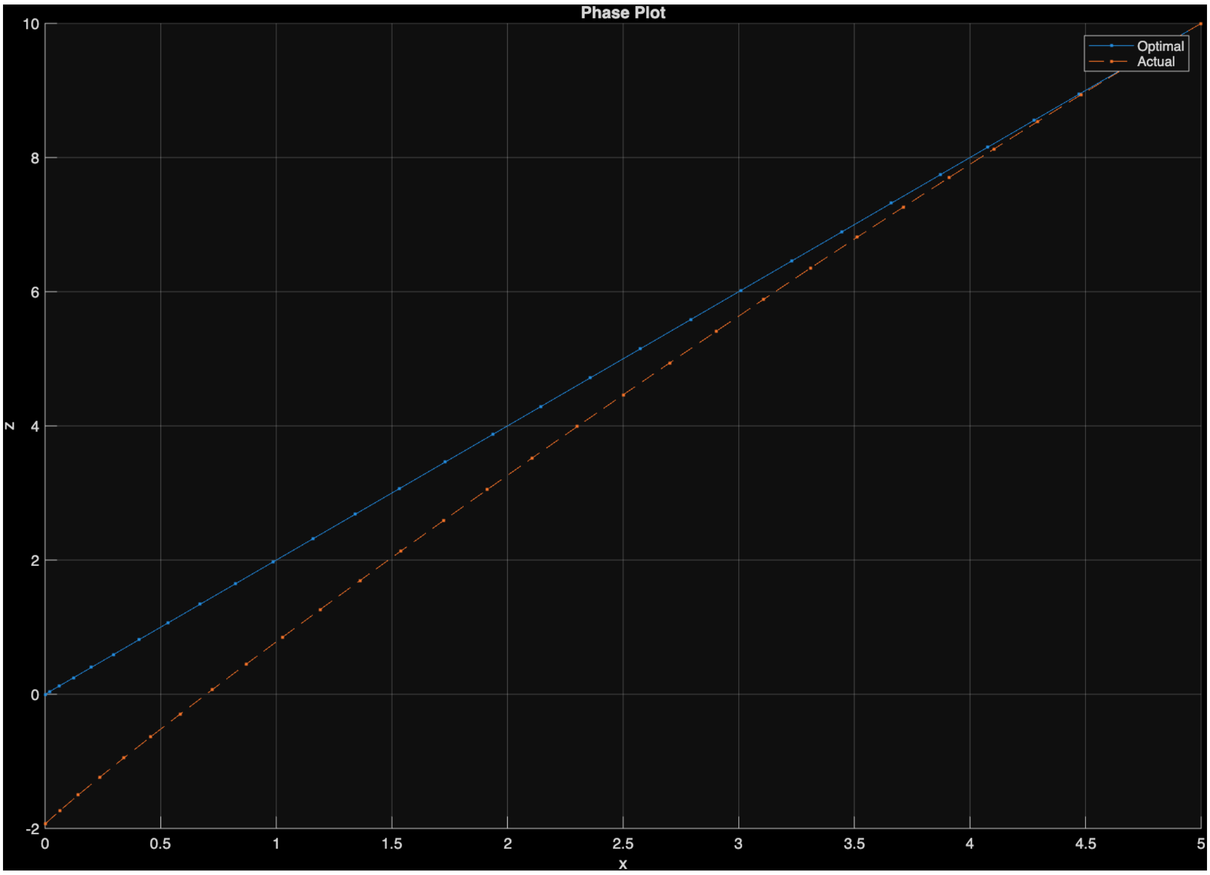


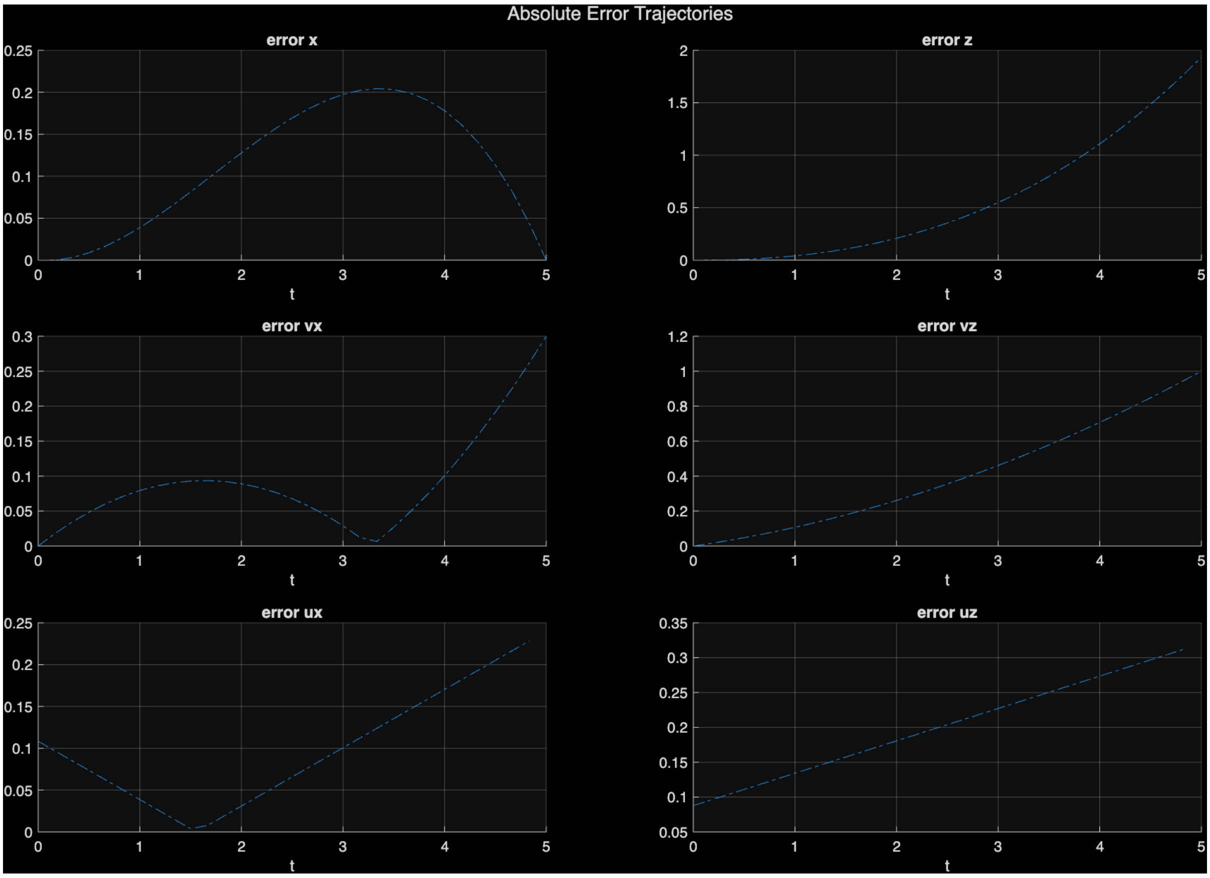
Figure 9: umax = 2 and T = 20

Looks like umax has more influence in determining the feasible region than the final time T.

3. Problem 22 - OpenLoop Execution




Phase Plot
Optimal
Actual
z
x


Absolute Error Trajectories
error x
error z
error vx
error vz
error ux
error uz
t

4. Problem 23 - MPC

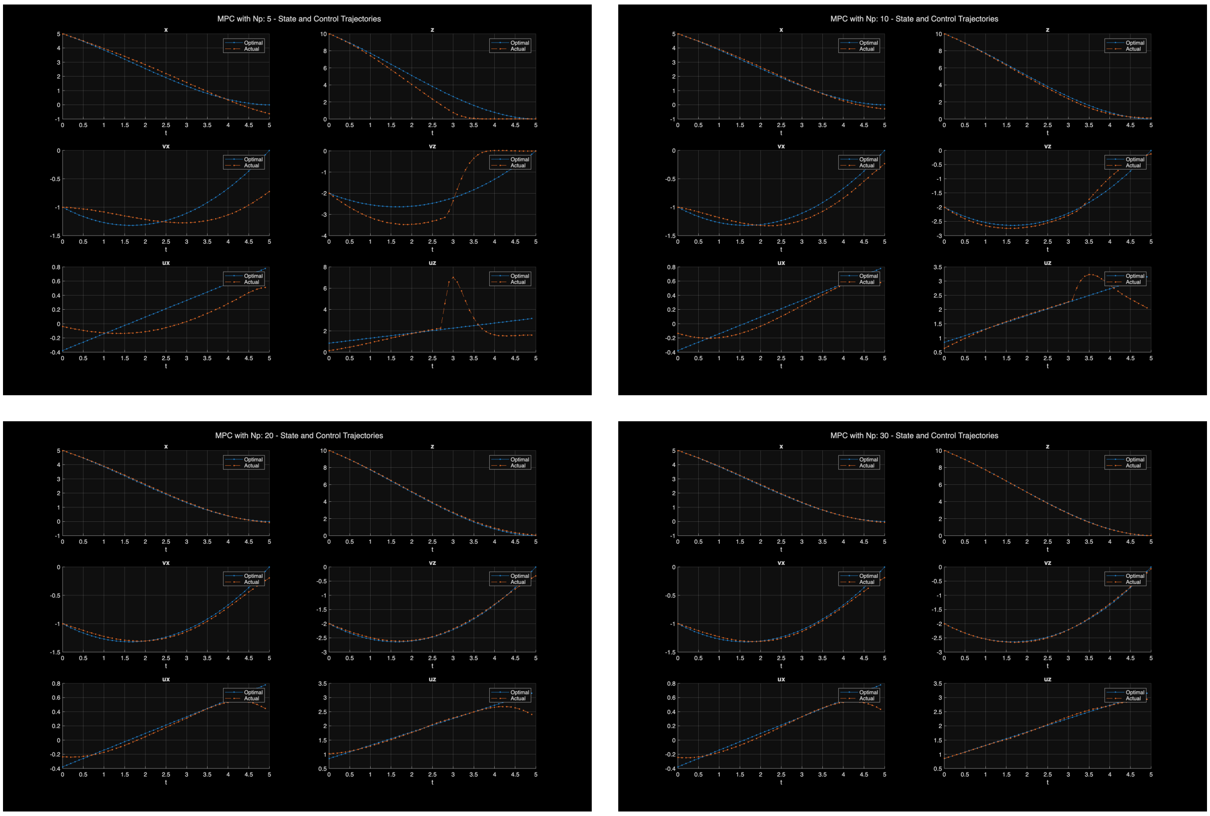


Figure 13: Trajectories for different Np

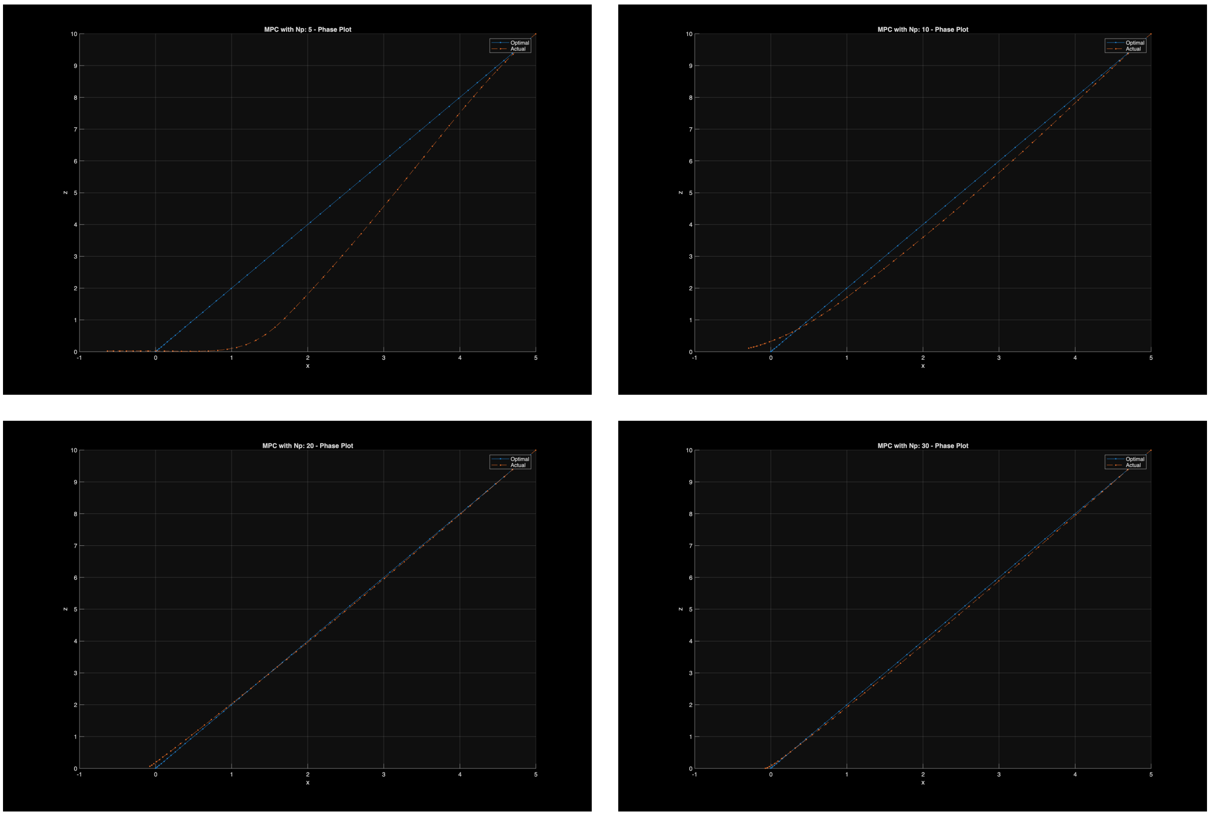


Figure 14: Phaseplots for different Np

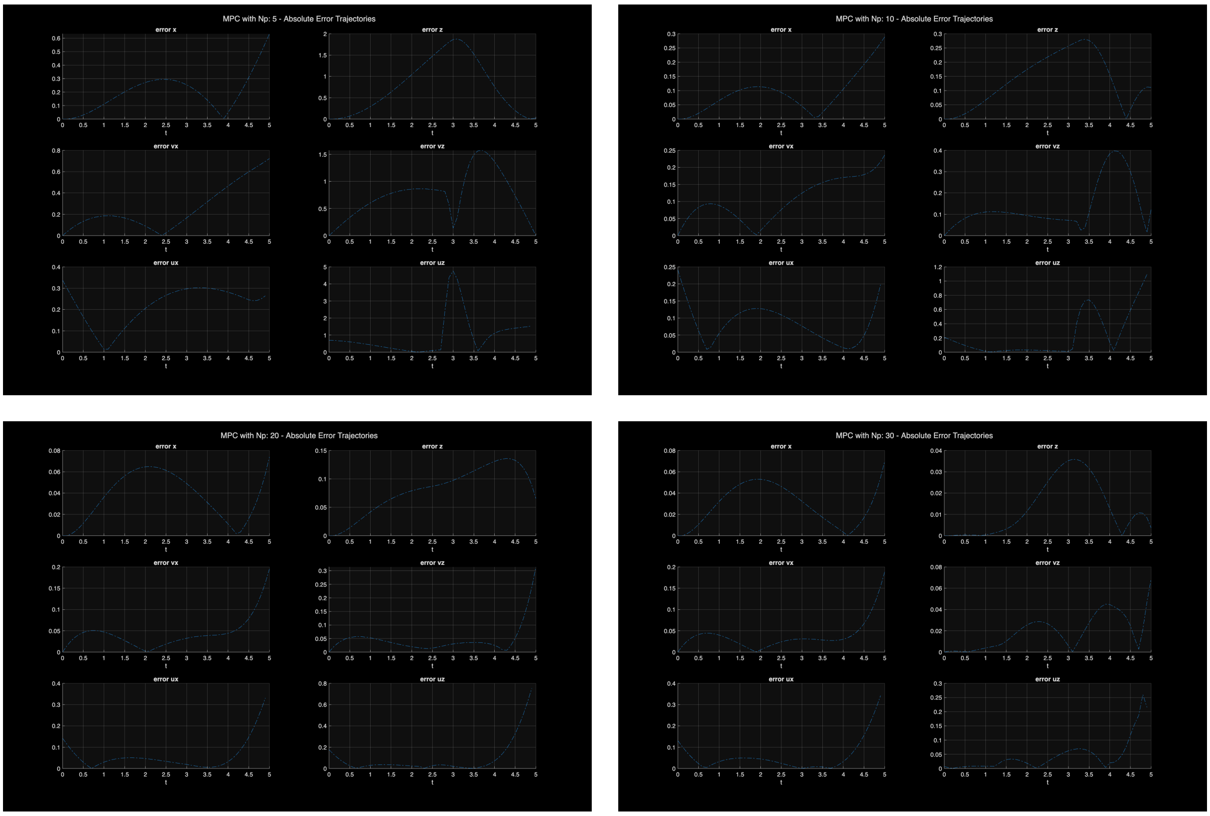


Figure 15: Errors for different Np